

Initiating Search

November 6, 2025, 7:35 PM

• Search:

CAS Lexicon Search:

Borane-tetrahydrofuran - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (1,715)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (869)	 Reactions	View Results
Filtered By:		
Document Type:	Journal	
Language:	English	

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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



31-513-CAS-16275481

Steps: 1 Yield: 100%

Synthesis of 1,3,6-Trisubstituted Azulenes Based on the 1-Acyloxyazulene Scaffold

By: Leino, Teppo O.; et al

European Journal of Organic Chemistry (2016), 2016(33), 5539-5544.

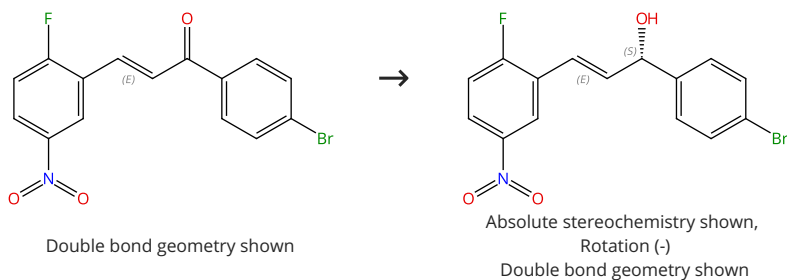
 1.1 Reagents: (7-4)-Trihydro(tetrahydrofuran)boron
 Solvents: Tetrahydrofuran; 20 min, rt

 1.2 Reagents: Sodium bicarbonate
 Solvents: Water

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



31-513-CAS-18789802

Steps: 1 Yield: 100%

Microwave-Assisted, Asymmetric Synthesis of 3-Amino-2,3-Dihydrobenzofuran Flavonoid Derivatives from Chalcones

By: Helgren, Travis R.; et al

Chemistry - A European Journal (2018), 24(18), 4509-4514.

 1.1 Reagents: (7-4)-Trihydro(tetrahydrofuran)boron
 Catalysts: (R)-2-Methyl-CBS-oxazaborolidine
 Solvents: Toluene, Tetrahydrofuran; 45 min, -20 °C

1.2 Solvents: Tetrahydrofuran; -20 °C; 1 h, -20 °C

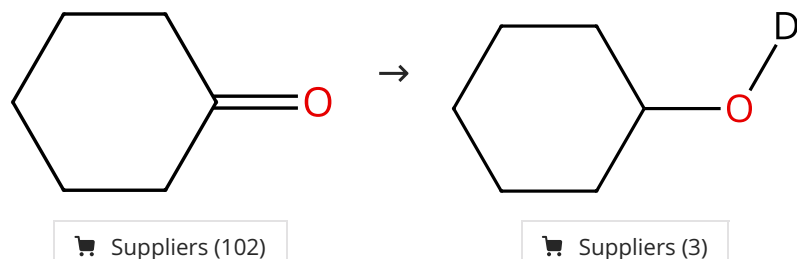
1.3 Solvents: Water; -20 °C → rt

 1.4 Reagents: Hydrochloric acid
 Solvents: Water; rt

Experimental Protocols

Scheme 3 (1 Reaction)

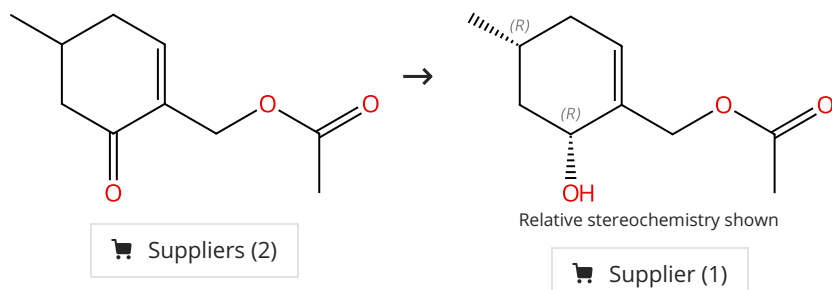
Steps: 1 Yield: 100%



31-513-CAS-9703286	Steps: 1 Yield: 100%	Ammonia borane as a metal free reductant for ketones and aldehydes: a mechanistic study By: Yang, Xianghua; et al Tetrahedron (2011), 67(37), 7121-7127.
1.1 Reagents: Ammonia borane Solvents: Methanol- <i>d</i> ₄ ; rt; > 1 min, rt		

Scheme 4 (1 Reaction)

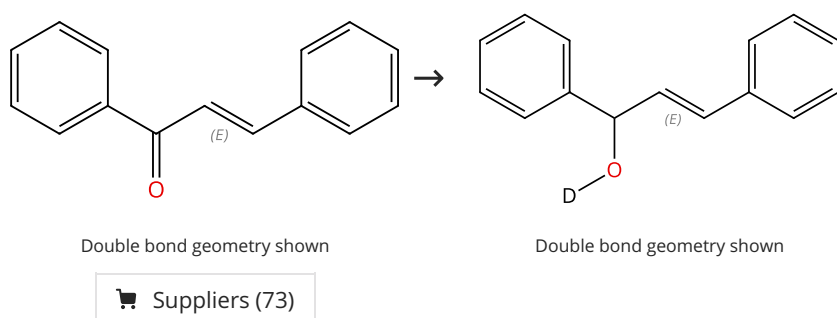
Steps: 1 Yield: 100%



31-513-CAS-4615187	Steps: 1 Yield: 100%	A formal total synthesis of lycopodium alkaloid, (±)-magellanine, by using the intramolecular Pauson-Khand reaction By: Ishizaki, Miyuki; et al Tetrahedron (2005), 61(16), 4053-4065.
1.1 Reagents: Cerium trichloride Catalysts: Sodium borohydride Solvents: Methanol; 1.5 h, 0 °C 1.2 Reagents: Hydrochloric acid Solvents: Water		

Scheme 5 (1 Reaction)

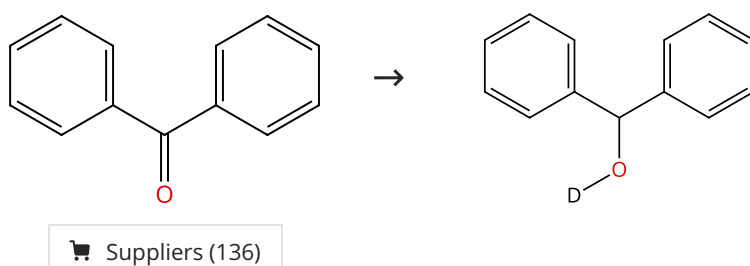
Steps: 1 Yield: 100%



31-513-CAS-5487718	Steps: 1 Yield: 100%	Ammonia borane as a metal free reductant for ketones and aldehydes: a mechanistic study By: Yang, Xianghua; et al Tetrahedron (2011), 67(37), 7121-7127.
1.1 Reagents: Ammonia borane Solvents: Methanol- <i>d</i> ₄ ; rt; > 1 min, rt		

Scheme 6 (1 Reaction)

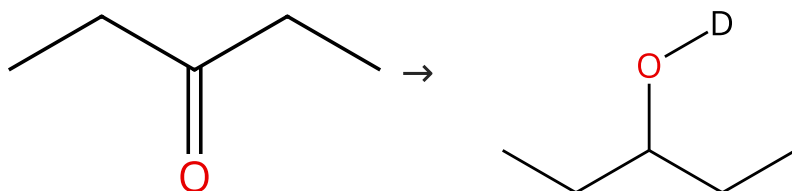
Steps: 1 Yield: 100%



31-513-CAS-984524	Steps: 1 Yield: 100%	Ammonia borane as a metal free reductant for ketones and aldehydes: a mechanistic study By: Yang, Xianghua; et al Tetrahedron (2011), 67(37), 7121-7127.
1.1 Reagents: Ammonia borane Solvents: Methanol- <i>d</i> ₄ ; rt; > 1 min, rt		

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%

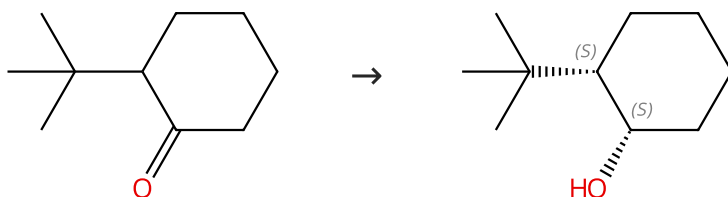


Suppliers (65)

31-513-CAS-7574889	Steps: 1 Yield: 100%	Ammonia borane as a metal free reductant for ketones and aldehydes: a mechanistic study By: Yang, Xianghua; et al Tetrahedron (2011), 67(37), 7121-7127.
1.1 Reagents: Ammonia borane Solvents: Methanol- <i>d</i> ₄ ; rt; > 1 min, rt		

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

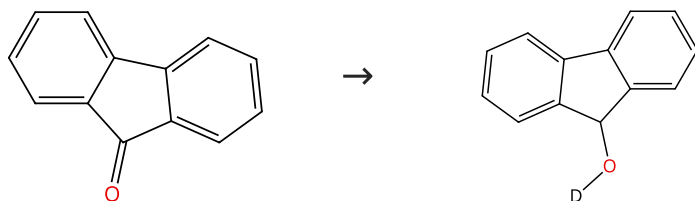
Suppliers (53)

Suppliers (9)

31-513-CAS-9872296	Steps: 1 Yield: 100%	Reaction of lithium (2,3-dimethyl-2-butyl)-t-butoxyborohydride with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2002), 23(6), 856-860.
1.1 Reagents: Borate(1-), dihydro(2-methyl-2-propanolato)(1,1,2-trimethylpropyl)-, lithium, (7-4)- Solvents: Pentane, Tetrahydrofuran		
1.2 Reagents: Sodium hydroxide, Hydrogen peroxide Solvents: Water		

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%

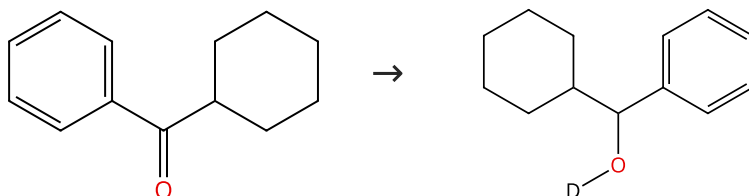


Suppliers (100)

31-513-CAS-1187147	Steps: 1 Yield: 100%	Ammonia borane as a metal free reductant for ketones and aldehydes: a mechanistic study By: Yang, Xianghua; et al Tetrahedron (2011), 67(37), 7121-7127.
1.1 Reagents: Ammonia borane Solvents: Methanol- <i>d</i> ₄ ; rt; > 1 min, rt		

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (80)

31-513-CAS-3302944	Steps: 1 Yield: 100%	Ammonia borane as a metal free reductant for ketones and aldehydes: a mechanistic study By: Yang, Xianghua; et al Tetrahedron (2011), 67(37), 7121-7127.
1.1 Reagents: Ammonia borane Solvents: Methanol- <i>d</i> ₄ ; rt; > 1 min, rt		

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (105)

Suppliers (9)

31-513-CAS-49324	Steps: 1 Yield: 100%	Reaction of lithium (2,3-dimethyl-2-butyl)-t-butoxyborohydride with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2002), 23(6), 856-860.
1.1 Reagents: Borate(1-), dihydro(2-methyl-2-propanolato)(1,1,2-trimethylpropyl)-, lithium, (7-4)- Solvents: Pentane, Tetrahydrofuran		
1.2 Reagents: Sodium hydroxide, Hydrogen peroxide Solvents: Water		

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

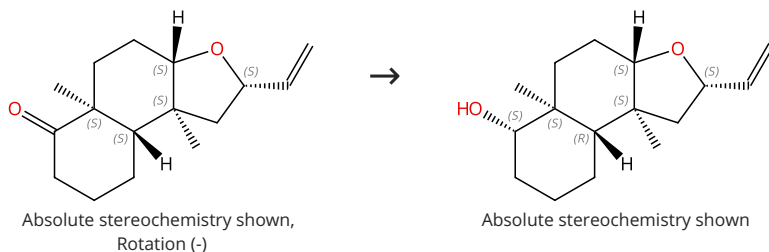
Suppliers (77)

Suppliers (42)

31-513-CAS-11955624 Steps: 1 Yield: 100% 1.1 Reagents: Borate(1-), dihydro(2-methyl-2-propanolato)(1,1,2-trimethylpropyl)-, lithium, (7-4)- Solvents: Pentane, Tetrahydrofuran 1.2 Reagents: Sodium hydroxide, Hydrogen peroxide Solvents: Water	Reaction of lithium (2,3-dimethyl-2-butyl)-t-butoxyborohydride with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2002), 23(6), 856-860.
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Scheme 13 (1 Reaction)

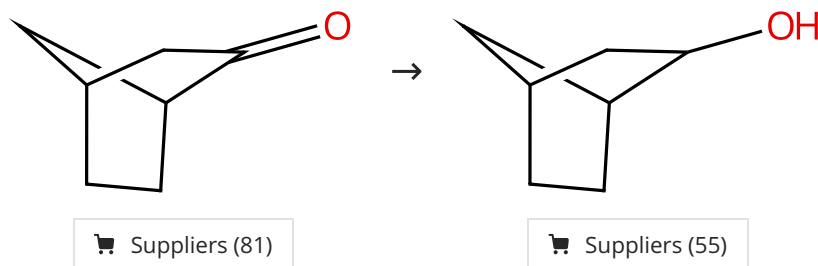
Steps: 1 Yield: 100%



31-513-CAS-20941129 Steps: 1 Yield: 100% 1.1 Reagents: Sodium borohydride Solvents: Ethanol; 30 min, 0 °C 1.2 Reagents: Ammonium chloride Solvents: Water Experimental Protocols	Synthesis and Biological Evaluation of Subglutininol Analogs for Immunomodulatory Agents By: Park, Hyeri; et al Journal of Medicinal Chemistry (2020), 63(1), 283-294.
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Scheme 14 (1 Reaction)

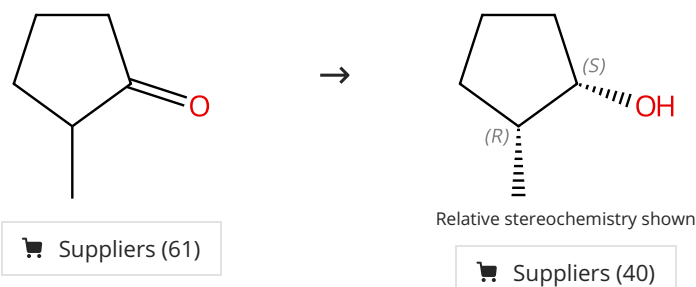
Steps: 1 Yield: 100%



31-513-CAS-10614876 Steps: 1 Yield: 100% 1.1 Reagents: (7-4)-Trihydro(tetrahydrofuran)boron Solvents: Tetrahydrofuran 1.2 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	A Solution of Borane in Tetrahydrofuran. A Stereoselective Reducing Agent for Reduction of Cyclic Ketones to Thermodynamically More Stable Alcohols By: Cha, Jin Soon; et al Journal of Organic Chemistry (2001), 66(22), 7514-7515.
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Scheme 15 (1 Reaction)

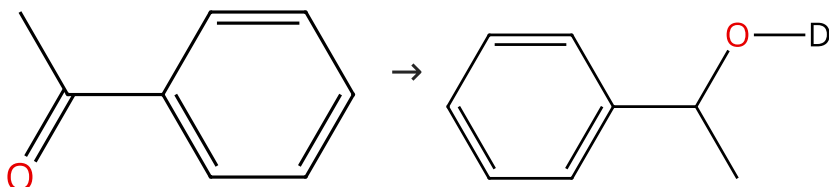
Steps: 1 Yield: 100%



31-513-CAS-12624319 Steps: 1 Yield: 100% 1.1 Reagents: Borate(1-), dihydro(2-methyl-2-propanolato)(1,1,2-trimethylpropyl)-, lithium, (7-4)- Solvents: Pentane, Tetrahydrofuran 1.2 Reagents: Sodium hydroxide, Hydrogen peroxide Solvents: Water	Reaction of lithium (2,3-dimethyl-2-butyl)-t-butoxyborohydride with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2002), 23(6), 856-860.
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Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



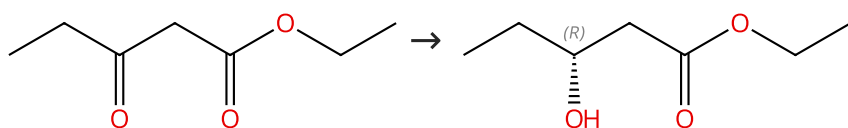
Suppliers (105)

Supplier (1)

31-513-CAS-14579687 Steps: 1 Yield: 100% 1.1 Reagents: Ammonia borane Solvents: Methanol-<i>d</i>₄; rt; > 1 min, rt	Ammonia borane as a metal free reductant for ketones and aldehydes: a mechanistic study By: Yang, Xianghua; et al Tetrahedron (2011), 67(37), 7121-7127.
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Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

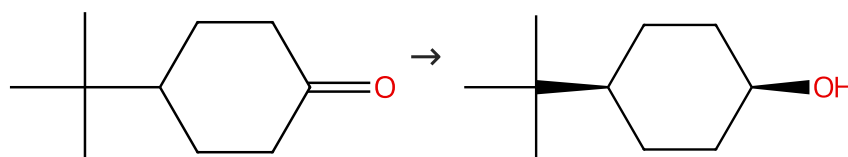
Suppliers (94)

Suppliers (18)

31-513-CAS-6074441 Steps: 1 Yield: 100% 1.1 Reagents: Hydrogen, Hydrochloric acid Catalysts: Di-μ-iodobis[(1,2,3,4,5,6-η)-1-methyl-4-(1-methyl ethyl)benzene]diruthenium, Ferrocene, 1,1''-methylenebis[2-(diphenylphosphino)-, (2<i>R</i>,2'<i>R</i>)- Solvents: Ethanol; 19 h, 80 bar, 25 °C	Preparation of C₂-symmetric bis[2-(diphenylphosphino) ferrocen-1-yl]-methane and its use in rhodium- and ruthenium-catalyzed hydrogenation By: Wang, Yaping; et al Helvetica Chimica Acta (2006), 89(8), 1772-1782.
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Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

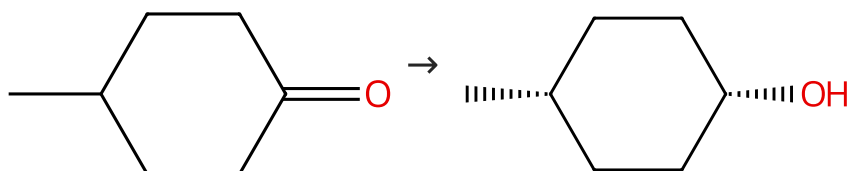
Suppliers (75)

Suppliers (57)

31-513-CAS-3419085	Steps: 1 Yield: 100%	Reaction of lithium (2,3-dimethyl-2-butyl)-t-butoxyborohydride with selected organic compounds containing representative functional groups
1.1 Reagents: Borate(1-), dihydro(2-methyl-2-propanolato)(1,1,2-trimethylpropyl)-, lithium, (7-4)- Solvents: Pentane, Tetrahydrofuran		By: Cha, Jin Soon; et al
1.2 Reagents: Sodium hydroxide, Hydrogen peroxide Solvents: Water		Bulletin of the Korean Chemical Society (2002), 23(6), 856-860.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

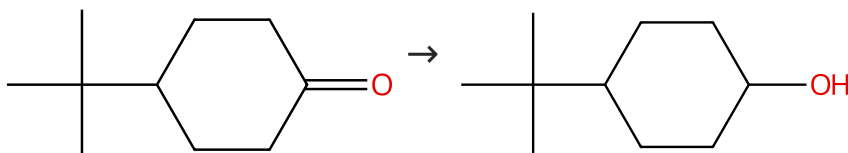
Suppliers (77)

Suppliers (54)

31-513-CAS-3133667	Steps: 1 Yield: 100%	Reaction of lithium (2,3-dimethyl-2-butyl)-t-butoxyborohydride with selected organic compounds containing representative functional groups
1.1 Reagents: Borate(1-), dihydro(2-methyl-2-propanolato)(1,1,2-trimethylpropyl)-, lithium, (7-4)- Solvents: Pentane, Tetrahydrofuran		By: Cha, Jin Soon; et al
1.2 Reagents: Sodium hydroxide, Hydrogen peroxide Solvents: Water		Bulletin of the Korean Chemical Society (2002), 23(6), 856-860.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



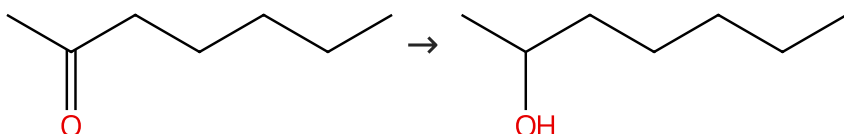
Suppliers (75)

Suppliers (78)

31-513-CAS-6348386	Steps: 1 Yield: 100%	A Solution of Borane in Tetrahydrofuran. A Stereoselective Reducing Agent for Reduction of Cyclic Ketones to Thermodynamically More Stable Alcohols
1.1 Reagents: (7-4)-Trihydro(tetrahydrofuran)boron Solvents: Tetrahydrofuran		By: Cha, Jin Soon; et al
1.2 Reagents: Hydrochloric acid Solvents: Water		Journal of Organic Chemistry (2001), 66(22), 7514-7515.
Experimental Protocols		

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



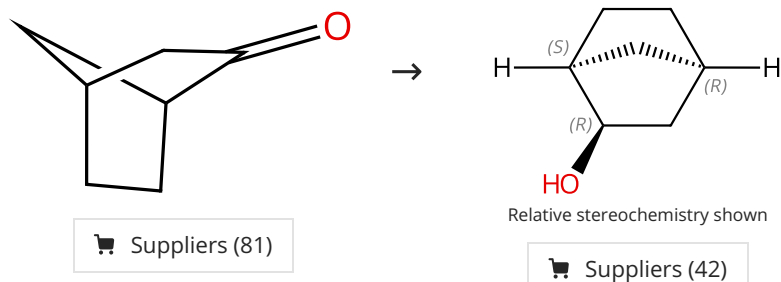
Suppliers (68)

Suppliers (77)

31-513-CAS-6584965 Steps: 1 Yield: 100%	Reaction of thexylalkoxyboranes with selected organic compounds containing representative functional groups. Comparison of reducing characteristics of the alkoxy derivatives By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (1998), 19(2), 243-249.
1.1 Reagents: Borinic acid, (1, 1,2-trimethylpropyl)-, ethyl ester Solvents: Tetrahydrofuran	

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



31-513-CAS-8721801 Steps: 1 Yield: 100%	Reaction of thexylalkoxyboranes with selected organic compounds containing representative functional groups. Comparison of reducing characteristics of the alkoxy derivatives By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (1998), 19(2), 243-249.
1.1 Reagents: Borinic acid, (1, 1,2-trimethylpropyl)-, ethyl ester Solvents: Tetrahydrofuran	

Scheme 23 (1 Reaction)

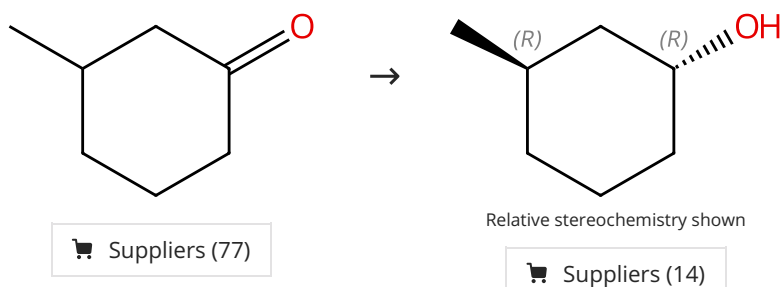
Steps: 1 Yield: 100%



31-513-CAS-8471084 Steps: 1 Yield: 100%	A Solution of Borane in Tetrahydrofuran. A Stereoselective Reducing Agent for Reduction of Cyclic Ketones to Thermodynamically More Stable Alcohols By: Cha, Jin Soon; et al Journal of Organic Chemistry (2001), 66(22), 7514-7515.
1.1 Reagents: (7-4)-Trihydro(tetrahydrofuran)boron Solvents: Tetrahydrofuran	
1.2 Reagents: Hydrochloric acid Solvents: Water	
Experimental Protocols	

Scheme 24 (1 Reaction)

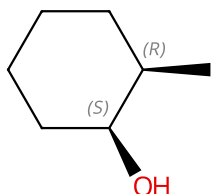
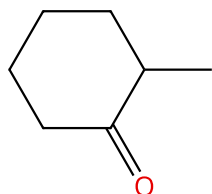
Steps: 1 Yield: 100%



31-513-CAS-8025422 Steps: 1 Yield: 100% 1.1 Reagents: Borate(1-), dihydro(2-methyl-2-propanolato)(1,1,2-trimethylpropyl)-, lithium, (7-4)- Solvents: Pentane, Tetrahydrofuran 1.2 Reagents: Sodium hydroxide, Hydrogen peroxide Solvents: Water	Reaction of lithium (2,3-dimethyl-2-butyl)-t-butoxyborohydride with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2002), 23(6), 856-860.
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Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

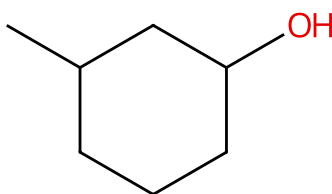
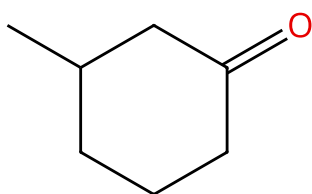
Suppliers (85)

Suppliers (39)

31-513-CAS-3465157 Steps: 1 Yield: 100% 1.1 Reagents: Borate(1-), dihydro(2-methyl-2-propanolato)(1,1,2-trimethylpropyl)-, lithium, (7-4)- Solvents: Pentane, Tetrahydrofuran 1.2 Reagents: Sodium hydroxide, Hydrogen peroxide Solvents: Water	Reaction of lithium (2,3-dimethyl-2-butyl)-t-butoxyborohydride with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2002), 23(6), 856-860.
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Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



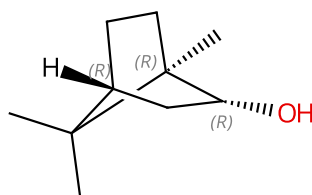
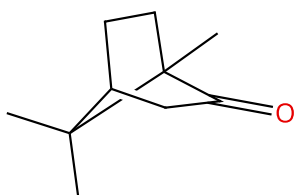
Suppliers (77)

Suppliers (51)

31-513-CAS-6062911 Steps: 1 Yield: 100% 1.1 Reagents: (7-4)-Trihydro(tetrahydrofuran)boron Solvents: Tetrahydrofuran 1.2 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	A Solution of Borane in Tetrahydrofuran. A Stereoselective Reducing Agent for Reduction of Cyclic Ketones to Thermodynamically More Stable Alcohols By: Cha, Jin Soon; et al Journal of Organic Chemistry (2001), 66(22), 7514-7515.
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Scheme 27 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

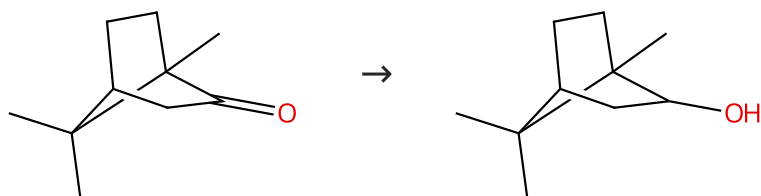
Suppliers (127)

Suppliers (73)

31-513-CAS-4953961 Steps: 1 Yield: 100% 1.1 Reagents: Borate(1-), dihydro(2-methyl-2-propanolato)(1,1,2-trimethylpropyl)-, lithium, (7-4)- Solvents: Pentane, Tetrahydrofuran 1.2 Reagents: Sodium hydroxide, Hydrogen peroxide Solvents: Water	Reaction of lithium (2,3-dimethyl-2-butyl)-t-butoxyborohydride with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2002), 23(6), 856-860.
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Scheme 28 (1 Reaction)

Steps: 1 Yield: 100%



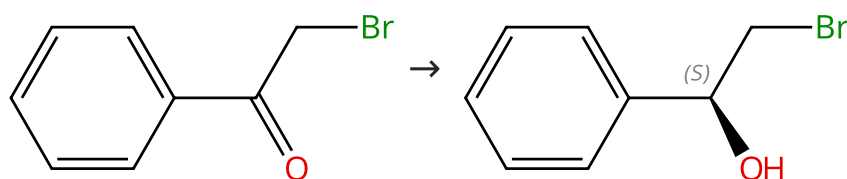
Suppliers (127)

Suppliers (34)

31-513-CAS-12738600 Steps: 1 Yield: 100% 1.1 Reagents: (7-4)-Trihydro(tetrahydrofuran)boron Solvents: Tetrahydrofuran 1.2 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	A Solution of Borane in Tetrahydrofuran. A Stereoselective Reducing Agent for Reduction of Cyclic Ketones to Thermodynamically More Stable Alcohols By: Cha, Jin Soon; et al Journal of Organic Chemistry (2001), 66(22), 7514-7515.
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Scheme 29 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (74)

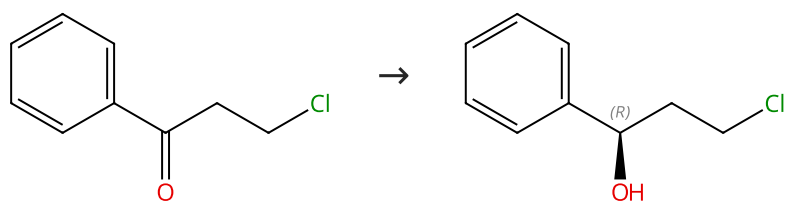
Absolute stereochemistry shown,
Rotation (+)

Suppliers (12)

31-513-CAS-3704481 Steps: 1 Yield: 100% 1.1 Reagents: (7-4)-Trihydro(tetrahydrofuran)boron Catalysts: Ferrocene, [(1<i>S</i>,2<i>S</i>)-2-amino-1-hydroxy-3,3-dimethylbutyl]- Solvents: Tetrahydrofuran; 50 min, 30 °C; 2 h, 30 °C 1.2 Reagents: Methanol 1.3 Reagents: Hydrochloric acid Solvents: Water	Chiral ferrocenyl amino alcohols as catalysts for the enantioselective borane reduction of ketones By: Brunin, Thierry; et al Tetrahedron: Asymmetry (2002), 13(12), 1241-1243.
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Scheme 30 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (89)

Absolute stereochemistry shown,
Rotation (+)

Suppliers (77)

31-513-CAS-405349

Steps: 1 Yield: 100%

Asymmetric synthesis of (R)-fluoxetine: a practical approach using recyclable and in-situ generated oxazaborolidine catalyst

By: Padiya, Kamlesh; et al

Chinese Journal of Chemistry (2009), 27(6), 1137-1140.

1.1 **Catalysts:** (7-4)-Trihydro(tetrahydrofuran)boron, α,α -Diphenyl-L-prolinol; 25 min, 28 °C; 15 min, 28 °C

1.2 **Solvents:** Toluene; 15 min, rt

1.3 **Reagents:** (7-4)-Trihydro(tetrahydrofuran)boron; 45 min, rt; 90 min, rt; cooled

Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (99)

Absolute stereochemistry shown,
Rotation (+)

Suppliers (69)

31-513-CAS-4363093

Steps: 1 Yield: 100%

Diphenylamine-derived bis-hydroxyamide catalyzed asymmetric borane reduction of prochiral ketones

By: Wang, Jin; et al

Tetrahedron: Asymmetry (2009), 20(5), 605-609.

1.1 **Reagents:** (7-4)-Trihydro(tetrahydrofuran)boron
Catalysts: 1,1'-(Iminodi-2,1-phenylene)bis[1-[(2S)-2-(hydroxy diphenylmethyl)-1-pyrrolidinyl]methanone]

Solvents: Dichloromethane; rt; rt → 50 °C; 1 h, 50 °C

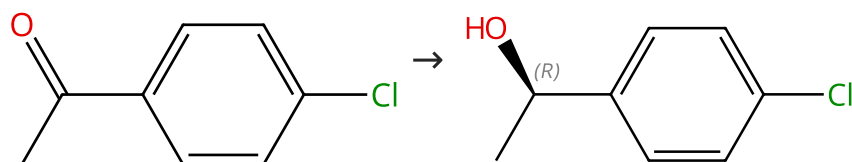
1.2 **Solvents:** Cyclohexane; 3 h, 50 °C

1.3 **Solvents:** Methanol

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (67)

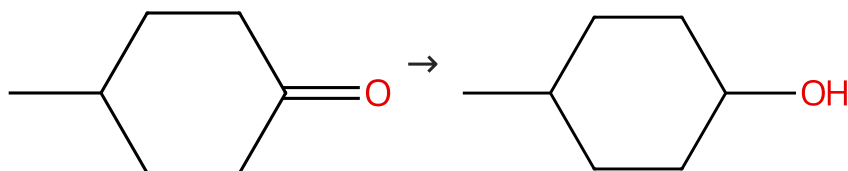
Absolute stereochemistry shown,
Rotation (+)

Suppliers (66)

31-513-CAS-14949383	Steps: 1 Yield: 100%	Chiral ferrocenyl amino alcohols as catalysts for the enantioselective borane reduction of ketones
1.1 Reagents: (7-4)-Trihydro(tetrahydrofuran)boron Catalysts: Ferrocene, [(1 <i>S</i> ,2 <i>S</i>)-2-amino-1-hydroxy-3,3-dimethylbutyl]- Solvents: Tetrahydrofuran; 50 min, 30 °C; 2 h, 30 °C		By: Brunin, Thierry; et al Tetrahedron: Asymmetry (2002), 13(12), 1241-1243.
1.2 Reagents: Methanol		
1.3 Reagents: Hydrochloric acid Solvents: Water		

Scheme 33 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (77)

Suppliers (73)

31-513-CAS-8199617	Steps: 1 Yield: 100%	A Solution of Borane in Tetrahydrofuran. A Stereoselective Reducing Agent for Reduction of Cyclic Ketones to Thermodynamically More Stable Alcohols
1.1 Reagents: (7-4)-Trihydro(tetrahydrofuran)boron Solvents: Tetrahydrofuran		By: Cha, Jin Soon; et al Journal of Organic Chemistry (2001), 66(22), 7514-7515.
1.2 Reagents: Hydrochloric acid Solvents: Water		
Experimental Protocols		

Scheme 34 (2 Reactions)

Steps: 1 Yield: 100%



Relative stereochemistry shown

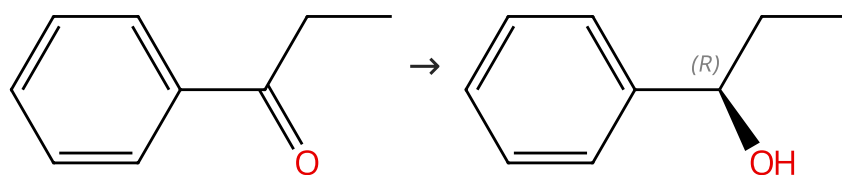
Suppliers (85)

Suppliers (50)

31-513-CAS-3928485	Steps: 1 Yield: 100%	A Solution of Borane in Tetrahydrofuran. A Stereoselective Reducing Agent for Reduction of Cyclic Ketones to Thermodynamically More Stable Alcohols
1.1 Reagents: (7-4)-Trihydro(tetrahydrofuran)boron Solvents: Tetrahydrofuran		By: Cha, Jin Soon; et al Journal of Organic Chemistry (2001), 66(22), 7514-7515.
1.2 Reagents: Hydrochloric acid Solvents: Water		
Experimental Protocols		
31-513-CAS-13394814	Steps: 1 Yield: 100%	A Solution of Borane in Tetrahydrofuran. A Stereoselective Reducing Agent for Reduction of Cyclic Ketones to Thermodynamically More Stable Alcohols
1.1 Reagents: Cyclohexanol, 2-methyl-, triester with boric acid (H ₃ BO ₃), (1 <i>R</i> ,1' <i>R</i> ,1'' <i>R</i> ,2 <i>S</i> ,2' <i>S</i> ,2'' <i>S</i>)- <i>rel</i> - Solvents: Tetrahydrofuran		By: Cha, Jin Soon; et al Journal of Organic Chemistry (2001), 66(22), 7514-7515.
1.2 Reagents: Hydrochloric acid Solvents: Water		
Experimental Protocols		

Scheme 35 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

Absolute stereochemistry shown,
Rotation (+)

Suppliers (68)

31-513-CAS-10684716

Steps: 1 Yield: 100%

Chiral ferrocenyl amino alcohols as catalysts for the enantioselective borane reduction of ketones

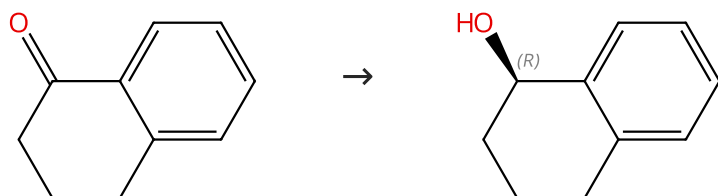
- 1.1 **Reagents:** (7-4)-Trihydro(tetrahydrofuran)boron
Catalysts: Ferrocene, [(1S,2S)-2-amino-1-hydroxy-3,3-dimethylbutyl]-
Solvents: Tetrahydrofuran; 50 min, 30 °C; 2 h, 30 °C
- 1.2 **Reagents:** Methanol
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Water

By: Brunin, Thierry; et al

Tetrahedron: Asymmetry (2002), 13(12), 1241-1243.

Scheme 36 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (99)

Absolute stereochemistry shown

Suppliers (78)

31-513-CAS-4278719

Steps: 1 Yield: 100%

Chiral ferrocenyl amino alcohols as catalysts for the enantioselective borane reduction of ketones

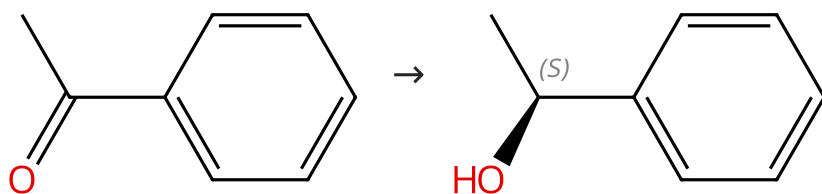
- 1.1 **Reagents:** (7-4)-Trihydro(tetrahydrofuran)boron
Catalysts: Ferrocene, [(1S,2S)-2-amino-1-hydroxy-3,3-dimethylbutyl]-
Solvents: Tetrahydrofuran; 50 min, 30 °C; 2 h, 30 °C
- 1.2 **Reagents:** Methanol
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Water

By: Brunin, Thierry; et al

Tetrahedron: Asymmetry (2002), 13(12), 1241-1243.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (105)

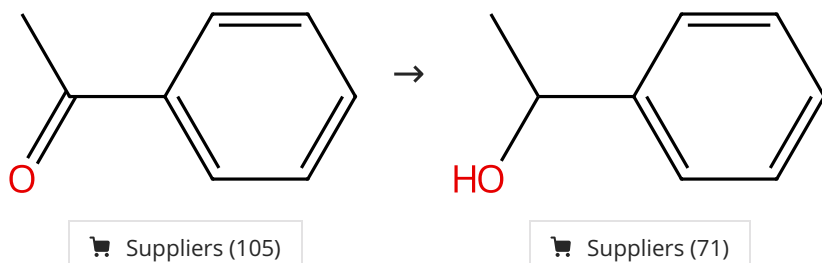
Absolute stereochemistry shown,
Rotation (-)

Suppliers (90)

31-513-CAS-6761076 Steps: 1 Yield: 100% 1.1 Catalysts: (Dimethyl sulfide)trihydroboron, (1<i>S</i>,2<i>R</i>,4<i>R</i>)-2-Amino-7,7-dimethylbicyclo[2.2.1]heptane-1-methanol Solvents: Tetrahydrofuran 1.2 Reagents: Hydrochloric acid Solvents: Water	New 1,3-amino alcohols derived from ketopinic acid and their application in catalytic enantioselective reduction of prochiral ketones By: Li, Xingshu; et al Tetrahedron: Asymmetry (1999), 10(4), 759-763.
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Scheme 38 (2 Reactions)

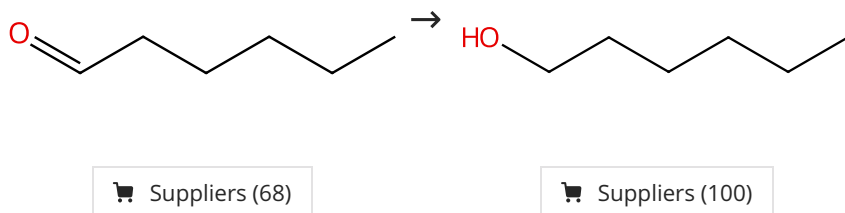
Steps: 1 Yield: 100%



31-513-CAS-1132529 Steps: 1 Yield: 100% 1.1 Reagents: Borate(1-), dihydro(2-methyl-2-propanolato)(1,1,2-trimethylpropyl)-, lithium, (7-4)- Solvents: Pentane, Tetrahydrofuran 1.2 Reagents: Sodium hydroxide, Hydrogen peroxide Solvents: Water	Reaction of lithium (2,3-dimethyl-2-butyl)-t-butoxyborohydride with selected organic compounds containing representative functional groups By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (2002), 23(6), 856-860.
31-513-CAS-10846947 Steps: 1 Yield: 100% 1.1 Reagents: Borinic acid, (1,1,2-trimethylpropyl)-, ethyl ester Solvents: Tetrahydrofuran	Reaction of hexylalkoxyboranes with selected organic compounds containing representative functional groups. Comparison of reducing characteristics of the alkoxy derivatives By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (1998), 19(2), 243-249.

Scheme 39 (1 Reaction)

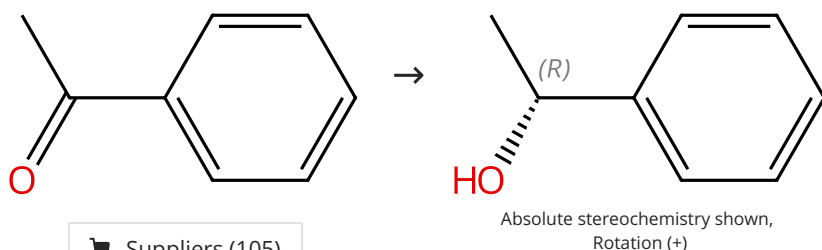
Steps: 1 Yield: 100%



31-513-CAS-2322545 Steps: 1 Yield: 100% 1.1 Reagents: Borinic acid, (1,1,2-trimethylpropyl)-, ethyl ester Solvents: Tetrahydrofuran	Reaction of hexylalkoxyboranes with selected organic compounds containing representative functional groups. Comparison of reducing characteristics of the alkoxy derivatives By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (1998), 19(2), 243-249.
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Scheme 40 (3 Reactions)

Steps: 1 Yield: 100%



Suppliers (105)

Suppliers (78)

31-513-CAS-12228161

Steps: 1 Yield: 100%

- 1.1 **Reagents:** (Dimethyl sulfide)trihydroboron
Catalysts: (3a*R*,8a*S*)-3,3a,8,8a-Tetrahydro-2-methyl-2*H*-indeno [1,2-*d*]-1,3,2-oxazaborole
Solvents: Tetrahydrofuran; 30 min, 23 °C
- 1.2 **Reagents:** Acetophenone
Solvents: Tetrahydrofuran; 30 min, rt
- 1.3 **Solvents:** Methanol; rt

Evaluating the role of solvent and borane on the enantioselectivity of the oxazaborolidine reduction of prochiral ketones using catalysts derived from cis-(1*R*,2*S*)-1-amino-indan-2-ol

By: Gilmore, Nathan J.; et al

Tetrahedron: Asymmetry (2003), 14(15), 2115-2118.

31-513-CAS-2161848

Steps: 1 Yield: 100%

- 1.1 **Reagents:** (*T*-4)-Trihydro(tetrahydrofuran)boron
Catalysts: Ferrocene, [(1*S*,2*S*)-2-amino-1-hydroxy-3,3-dimethylbutyl]-
Solvents: Tetrahydrofuran; 50 min, 30 °C; 2 h, 30 °C
- 1.2 **Reagents:** Methanol
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Water

Chiral ferrocenyl amino alcohols as catalysts for the enantioselective borane reduction of ketones

By: Brunin, Thierry; et al

Tetrahedron: Asymmetry (2002), 13(12), 1241-1243.

31-513-CAS-15397632

Steps: 1 Yield: 100%

- 1.1 **Reagents:** (*T*-4)-Trihydro(tetrahydrofuran)boron
Catalysts: (*S*)-1,2,3,4-Tetrahydro-3-isoquinolinemethanol
Solvents: Tetrahydrofuran

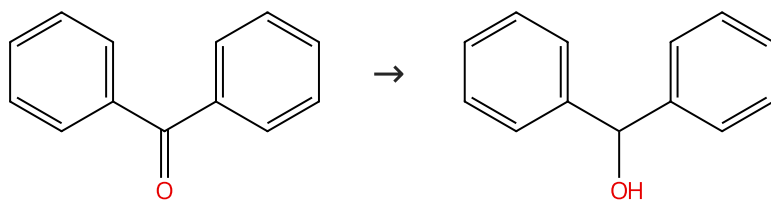
Enantioselective catalytic borane reductions of aromatic ketones: syntheses and application of two chiral β -amino alcohols from (*S*)-porretine

By: Stingl, Klaus; et al

Tetrahedron: Asymmetry (1992), 3(2), 223-6.

Scheme 41 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (136)

Suppliers (97)

31-513-CAS-7509152

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Borate(1-), dihydro(2-methyl-2-propanolato)(1,1,2-trimethylpropyl)-, lithium, (*T*-4)-
Solvents: Pentane, Tetrahydrofuran
- 1.2 **Reagents:** Sodium hydroxide, Hydrogen peroxide
Solvents: Water

Reaction of lithium (2,3-dimethyl-2-butyl)-*t*-butoxyborohydride with selected organic compounds containing representative functional groups

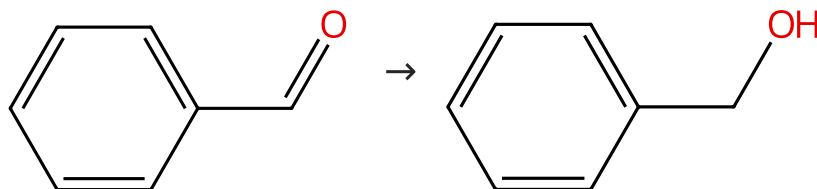
By: Cha, Jin Soon; et al

Bulletin of the Korean Chemical Society (2002), 23(6), 856-860.

31-513-CAS-12982197 Steps: 1 Yield: 100% 1.1 Reagents: Borinic acid, (1, 1,2-trimethylpropyl)-, ethyl ester Solvents: Tetrahydrofuran	Reaction of thexylalkoxyboranes with selected organic compounds containing representative functional groups. Comparison of reducing characteristics of the alkoxy derivatives By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (1998), 19(2), 243-249.
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Scheme 42 (3 Reactions)

Steps: 1 Yield: 100%



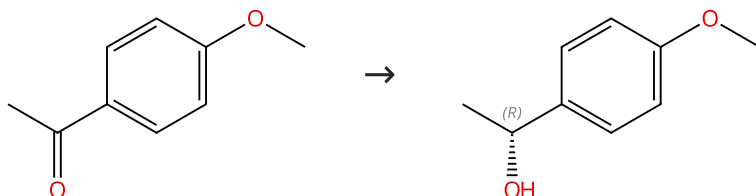
Suppliers (69)

Suppliers (148)

31-513-CAS-11832855 Steps: 1 Yield: 100% 1.1 Reagents: Ammonia borane Solvents: Methanol-<i>d</i>₄; rt; > 1 min, rt	Ammonia borane as a metal free reductant for ketones and aldehydes: a mechanistic study By: Yang, Xianghua; et al Tetrahedron (2011), 67(37), 7121-7127.
31-513-CAS-5723147 Steps: 1 Yield: 100% 1.1 Reagents: Ammonia borane Solvents: Methanol; < 15 min, rt	Ammonia borane as a metal free reductant for ketones and aldehydes: a mechanistic study By: Yang, Xianghua; et al Tetrahedron (2011), 67(37), 7121-7127.
31-513-CAS-4453220 Steps: 1 Yield: 100% 1.1 Reagents: Borinic acid, (1, 1,2-trimethylpropyl)-, ethyl ester Solvents: Tetrahydrofuran	Reaction of thexylalkoxyboranes with selected organic compounds containing representative functional groups. Comparison of reducing characteristics of the alkoxy derivatives By: Cha, Jin Soon; et al Bulletin of the Korean Chemical Society (1998), 19(2), 243-249.

Scheme 43 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (98)

Absolute stereochemistry shown,
Rotation (+)

Suppliers (52)

31-513-CAS-4532106	Steps: 1 Yield: 99%	Novel fluororous prolinol as a pre-catalyst for catalytic asymmetric borane reduction of various ketones
1.1 Catalysts: (7-4)-Trihydro(tetrahydrofuran)boron, (2,5)-α,α-Bis (3,3,4,4,5,5,6,6,7,7,8,8,8-tridecafluorooctyl)-2-pyrrolidin emethanol Solvents: Tetrahydrofuran, Toluene-d_8; 1 h, reflux; reflux $\rightarrow$ rt 1.2 Reagents: (7-4)-Trihydro(tetrahydrofuran)boron Solvents: Tetrahydrofuran, Toluene-d_8; 3 h, rt; 1 h, rt 1.3 Reagents: Methanol, Sodium bicarbonate Solvents: Methanol, Water	By: Goushi, Sakiko; et al Tetrahedron (2007), 63(19), 4061-4066.	
Experimental Protocols		